INCH-POUND
MIL-DTL-24202A
w/AMENDMENT 1
18 May 2021
SUPERSEDING
MIL-DTL-24202A
17 October 2017

DETAIL SPECIFICATION

FITTINGS, BUTT-WELDING, SEAMLESS OR WELDED, 70-30 COPPER-NICKEL ALLOY, 700 PSI, 200 °F

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 70-30 copper-nickel alloy seamless or welded butt-welding fittings with a pressure rating of 700 pounds per square inch (PSI) at 200 °F in sizes ½-inch up to and including 16-inch nominal pipe size (NPS).

1.2 Classification. Fittings are of the following types, as specified (see 6.2):

a. Type I - Seamless

b. Type II - Welded

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-15726	-	Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod, and Wire
MIL-T-16420	-	Tube, Copper-Nickel Alloy, Seamless and Welded (Copper Alloy Numbers 715 and 706)
MIL-C-24679	-	Copper-Nickel Alloy Forgings and Forging Stock

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4730

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

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S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.9 - Factory-Made Wrought Buttwelding Fittings

(Copies of this document are available online at www.asme.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials and processing.

3.2.1 Methods. Fittings shall be made from pipe, tubing, forgings, bar stock, or plates by extruding, forming, forging, machining, fusion welding, or by a combination of two or more of these operations.

3.2.2 Machining. Fittings machined from forged bars are acceptable. Fittings machined directly from rolled bars are not acceptable unless at least ⅓ of the diameter of the bar as rolled is removed from the center of the bar in the machining process.

3.2.3 Fittings made from pipe or tubing. Fittings made from pipe or tubing shall meet the chemical and physical requirements, with the exception of the hardness requirement, of 70-30 copper-nickel alloy in accordance with MIL-T-16420.

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3.2.4 Fittings made from bar stock, plates, or forgings. Fittings made from bar stock, plates, or forgings shall meet the chemical and physical requirements of 70-30 copper-nickel alloy in accordance with MIL-C-15726 or MIL-C-24679, except as modified herein.

3.2.5 Annealing. All finished fittings shall be furnished in the fully annealed condition. Annealing shall be performed at 1,300±50 °F per hour per inch of thickness.

3.3 Design.

3.3.1 General. All fittings furnished under this specification shall be in accordance with ASME B16.9 and the additional requirements of 3.3.2, with the exception of materials, which shall comply with 3.2.

3.3.2 Caps. Caps shall be ellipsoidal in profile and shall be in accordance with ASME B16.9 and herein, with the exception of materials, which shall comply with 3.2.

3.4 Dimensions.

3.4.1 Measurement system. Unless otherwise specified herein, all dimensions are in inches.

3.4.2 Fitting dimensions. Fitting dimensions shall be in accordance with figures 1 through 15 and tables I through VI.

3.4.3 Minimum wall thickness. At no point throughout the fitting shall the wall thickness be less than dimension "T" shown on figure 2 (see table I). For reducing fittings, the minimum wall thickness at any location shall be based on the external diameter at that location.

3.4.4 Tangents. All elbows and return bends shall have a tangent on each end in accordance with column D of table I.

3.4.4.1 Consumable insert fitting. When fittings are ordered for consumable insert welding (see 6.2), the tangent may be omitted. The center-to-end dimension for fittings intended for consumable insert welding shall be as shown in table I less the length of the tangent.

3.4.5 Dimensional tolerances. Tolerances on dimensions shall be in accordance with table II.

3.4.6 Dimensional and contour changes. Any modifications to fitting design involving dimensional or contour changes, other than weld end preparation, shall require a retest in accordance with 4.2. prior to acceptance.

3.5 Fabrication.

3.5.1 Welding. All welding shall be in accordance with S9074-AR-GIB-010/278.

3.5.2 Number of welds. Fittings shall contain no more than one weld, either longitudinal or circumferential. The weld shall not be located in any of the principal axes as shown on figure 1.

3.5.3 Weld end details. Unless otherwise specified (see 6.2), all fittings shall be furnished with weld end details in accordance with figure 2.

3.6 Marking.

3.6.1 Legibility and permanence. Each fitting shall be legibly and permanently marked in accordance with 3.6.2 and 3.6.3.

3.6.2 Content. Marking shall include manufacturer's name or trademark, size (NPS) 700 WOG[1], type I (for seamless) or type II (for welded), manufacturer's heat code identification number, and MIL-DTL-24202.

3.6.3 Size. The required markings shall be ⅛ to ½ inch high, depending on the size of the fitting.

[1] "WOG" is a common industry acronym for valves, such as in this specification, suitable for "Water, Oil, or Gas".

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the fittings when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5 and 4.6.

4.2.1 Fittings with wall thickness shown in tables I through IV. One sample fitting of each type and each size and wall thickness shown in tables I through IV shall be burst tested in accordance with the requirements of ASME B16.9. In the case of reducing fittings, a satisfactory test of a straight fitting that is the same size as the largest connection of the reducing fitting shall eliminate the need of testing the reducing fittings. Failure of associated test equipment, such as connecting piping below the burst test pressure, shall require retesting.

4.2.2 Fittings with wall thickness shown in tables V and VI. One 4-inch sample tee fitting with the dimensions shown in tables I through IV and the wall thickness shown in tables V and VI shall be satisfactorily burst tested in accordance with the requirements of ASME B16.9.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.1.

4.3.1 Lot. All fittings of the same type and size offered for delivery at one time shall be considered a lot for the purpose of sampling.

4.3.2 Sampling for visual and dimensional examination. A random sample of fittings shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.4 Visual and dimensional examination. Each of the sample fittings selected in accordance with 4.3.2 shall be visually and dimensionally examined to verify compliance with all the requirements of this specification not involving tests. As a minimum, each sample fitting shall be examined for adequate thickness at the locations shown on figures 11 through 15 and delineated by "X". Each section so marked shall have the thickness checked at four locations around the circumference, 90 degrees apart, except for the measurement of the crotches. For tee crotches, the measurement shall be taken as shown on figure 13. The numbers shown on figure 13 indicate the number and location of inspection points.

4.5 Nondestructive inspection.

4.5.1 Dye penetrant inspection. Dye penetrant inspection shall be performed in accordance with S9074-AR-GIB-010/278. Dye penetrant inspection shall be conducted as follows:

a. Butt-weld end preparation - All sizes

b. External surfaces - All sizes

c. Internal surfaces - 2 inches and above.

All defects in butt-weld preparation found when dye penetrant is inspected shall be removed prior to acceptance. Only linear defects, as defined in S9074-AR-GIB-010/278, shall be removed from other surfaces prior to acceptance. Repair welding shall be in accordance with S9074-AR-GIB-010/278.

4.5.2 Radiographic inspection. All welds shall be 100 percent radiographed in accordance with S9074-AR-GIB-010/278. Acceptance criteria shall be in accordance with MIL-STD-2035.

4.6 Material control. DELETED

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Fittings covered by this specification are intended primarily for naval shipboard water and gas service. Existing stock of fittings manufactured to MIL-DTL-24202A dated 17 October 2017 is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. When first article inspection is required (see 3.1).
d. Whether tangents may be omitted when used for consumable insert welding (see 3.4.4.1).
e. Weld end details if not as specified herein (see 3.5.3).
f. AQL of 1.0 (see 4.3.2).
g. DELETED
h. Packaging requirements (see 5.1).
i. Size required (see tables I through VI).
j. Drawings required.

6.3 Subject term (key word) listing.

Annealing
Bend, return
Reducer
Tangent

6.4 DELETED

6.4.1 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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TABLE I. General dimensions.

NPS	Pipe O.D. 1/	T	F	D	A		B	C	E	P	K
		Minimum 'C' bore thickness	Minimum counterbore	Minimum tangent	Center to end of 90-degree elbows		Center to end of 45-degree elbows long radius	Tees straight	Caps	180-degree return	180-degree return
					Long radius	Short radius					
½	0.840	0.072	5/16	5/16	1½	--	5/8	--	--	--	--
¾	1.050	0.083	5/16	5/16	1½	--	¾	1⅛	--	--	--
1	1.315	0.095	5/16	5/16	1½	1	⅞	1½	1½	3	2½
1¼	1.660	0.095	5/16	5/16	1⅞	1¼	1	1⅞	1½	3¾	3 1/16
1½	1.900	0.109	5/16	5/16	2¼	1⅕	1⅛	2¼	1½	4½	3 9/16
2	2.375	0.120	5/16	5/16	3	2	1 3/16	2½	1½	6	4½
2½	2.875	0.134	5/16	5/16	3¾	2½	1¾	3	1½	7½	5½
3	3.500	0.165	5/16	5/16	4½	3	2	3⅜	2	9	6 9/16
3½	4.000	0.180	½	½	5¼	3½	2¼	3¾	2½	10½	7¾
4	4.500	0.203	½	½	6	4	2½	4⅛	2½	12	8¾
5	5.563	0.220	½	½	7½	5	3⅛	4⅞	3	15	10 13/16
6	6.625	0.259	½	½	9	6	3¾	5⅝	3½	18	12 13/16
8	8.625	0.340	½	½	12	8	5	7	4	24	16 13/16
10	10.750	0.380	½	½	15	10	6¼	8½	5	30	20⅞
12	12.750	0.454	½	½	18	12	7½	10	6	36	24⅞
14	14.000	0.473	½	½	21	14	8¾	11	6½	42	28½
16	16.000	0.534	½	½	24	16	10	12	7	48	32½

FOOTNOTE:

1/ "O.D." stands for "outside diameter".

NOTE: All dimensions are in inches.

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TABLE II. Tolerances.

NPS	A, B, C, M, and H ±	E ±	P ±	K ±	U ±
½ to 4½	1/16	1/8	¼	¼	1/32
5 to 9	1/16	¼	¼	¼	1/32
10 to 16	1/32	¼	⅜	¼	1/16
NOTE: All dimensions are in inches.					

TABLE III. Concentric and eccentric reducer dimensions.

NPS	H
1 × ¾	2
1 × ½	2
1¼ × 1	2
1¼ × ¾	2
1¼ × ½	2
1½ × 1¼	2½
1½ × 1	2½
1½ × ¾	2½
1½ × ½	2½
2 × 1½	3
2 × 1¼	3
2 × 1	3
2 × ¾	3
2½ × 2	3½
2½ × 1½	3½
2½ × 1¼	3½
2½ × 1	3½
3 × 2½	3½
3 × 2	3½
3 × 1½	3½
3 × 1¼	3½
3½ × 3	4
3½ × 2½	4
3½ × 2	4
3½ × 1½	4
3½ × 1¼	4
4 × 3½	4

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TABLE III. Concentric and eccentric reducer dimensions – Continued.

NPS	H
4 × 3	4
4 × 2½	4
4 × 2	4
4 × 1½	4
5 × 4	5
5 × 3½	5
5 × 3	5
5 × 2½	5
5 × 2	5
6 × 5	5½
6 × 4	5½
6 × 3½	5½
6 × 3	5½
6 × 2½	5½
8 × 6	6
8 × 5	6
8 × 4	6
8 × 3½	6
10 × 8	7
10 × 6	7
10 × 5	7
10 × 4	7
12 × 10	8
12 × 8	8
12 × 6	8
12 × 5	8
14 × 12	13
14 × 10	13
14 × 8	13
14 × 6	13
16 × 14	14
16 × 12	14
16 × 10	14
16 × 8	14
NOTE: All dimensions are in inches.	

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TABLE IV. Reducing tee dimensions.

NPS	M
1 × 1 × ¾	1½
1 × 1 × ½	1½
1¼ × 1¼ × 1	1⅞
1¼ × 1¼ × ¾	1⅞
1¼ × 1¼ × ½	1⅞
1½ × 1½ × 1¼	2¼
1½ × 1½ × 1	2¼
1½ × 1½ × ¾	2¼
1½ × 1½ × ½	2¼
2 × 2 × 1½	2⅜
2 × 2 × 1¼	2¼
2 × 2 × 1	2
2 × 2 × ¾	1¾
2½ × 2½ × 2	2¾
2½ × 2½ × 1½	2⅝
2½ × 2½ × 1¼	2½
2½ × 2½ × 1	2¼
3 × 3 × 2½	3¼
3 × 3 × 2	3
3 × 3 × 1½	2⅞
3 × 3 × 1¼	2¾
3½ × 3½ × 3	3⅝
3½ × 3½ × 2½	3½
3½ × 3½ × 2	3¼
3½ × 3½ × 1½	3⅛
4 × 4 × 3½	4
4 × 4 × 3	3⅞
4 × 4 × 2½	3¾
4 × 4 × 2	3½
4 × 4 × 1½	3⅜
5 × 5 × 4	4⅝
5 × 5 × 3½	4½
5 × 5 × 3	4⅜
5 × 5 × 2½	4¼
5 × 5 × 2	4⅛
6 × 6 × 5	5⅜
6 × 6 × 4	5⅛

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TABLE IV. Reducing tee dimensions – Continued.

NPS	M
6 × 6 × 3½	5
6 × 6 × 3	4⅞
6 × 6 × 2½	4¾
8 × 8 × 6	6⅝
8 × 8 × 5	6⅜
8 × 8 × 4	6⅛
8 × 8 × 3½	6
10 × 10 × 8	8
10 × 10 × 6	7⅝
10 × 10 × 5	7½
10 × 10 × 4	7¼
12 × 12 × 10	9½
12 × 12 × 8	9
12 × 12 × 6	8⅝
12 × 12 × 5	8½
14 × 14 × 12	10⅝
14 × 14 × 10	10⅛
14 × 14 × 8	9¾
14 × 14 × 6	9⅜
16 × 16 × 14	12
16 × 16 × 12	11⅝
16 × 16 × 10	11⅛
16 × 16 × 8	10¾
16 × 16 × 6	10⅜
NOTE: All dimensions are in inches.	

TABLE V. Straight and reducing tee crotch thicknesses.

Size run × branch	Crotch minimum
2 × 2	0.190
2 × 1½	0.179
2 × 1¼	0.169
2 × 1	0.162
2 × ¾	0.157
2½ × 2½	0.218
2½ × 2	0.204
2½ × 1½	0.189
2½ × 1¼	0.185

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TABLE V. Straight and reducing tee crotch thicknesses – Continued.

Size run × branch	Crotch minimum
2½ × 1	0.179
3 × 3	0.280
3 × 2½	0.250
3 × 2	0.239
3 × 1½	0.229
3 × 1¼	0.224
3 × 1	0.216
4 × 4	0.356
4 × 3	0.318
4 × 3½	0.333
4 × 2½	0.296
4 × 2	0.285
4 × 1½	0.275
4 × 1¼	0.269
5 × 5	0.390
5 × 4	0.373
5 × 3½	0.350
5 × 3	0.335
5 × 2½	0.315
5 × 2	0.304
6 × 6	0.468
6 × 5	0.429
6 × 4	0.412
6 × 3½	0.390
6 × 3	0.378
6 × 2½	0.363
6 × 2	0.351
8 × 8	0.630
8 × 6	0.549
8 × 5	0.523
8 × 4	0.496
8 × 3½	0.483
8 × 3	0.472
10 × 10	0.710
10 × 8	0.670
10 × 6	0.589
10 × 5	0.565

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TABLE V. Straight and reducing tee crotch thicknesses – Continued.

Size run × branch	Crotch minimum
10 × 4	0.540
12 × 12	0.825
12 × 10	0.805
12 × 8	0.760
12 × 6	0.680
12 × 5	0.660
NOTE: All dimensions are in inches.	

TABLE VI. Elbow and return bend crotch thickness.

Size	Thickness	
	Short radius	Long radius
½	0.085	0.080
¾	0.093	0.087
1	0.104	0.096
1¼	0.118	0.108
1½	0.128	0.117
2	0.148	0.133
2½	0.168	0.151
3	0.194	0.173
3½	0.214	0.190
4	0.235	0.208
5	0.277	0.245
6	0.327	0.282
8	0.405	0.352
10	0.482	0.426
12	0.572	0.496
14	0.623	0.540
16	0.708	0.610
NOTE: All dimensions are in inches.		

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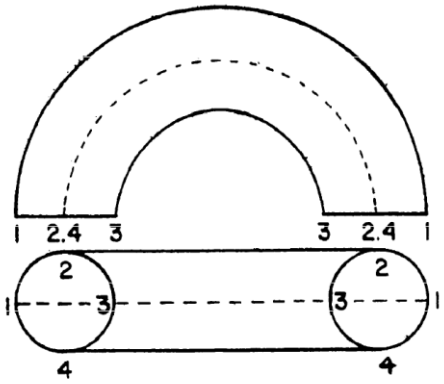
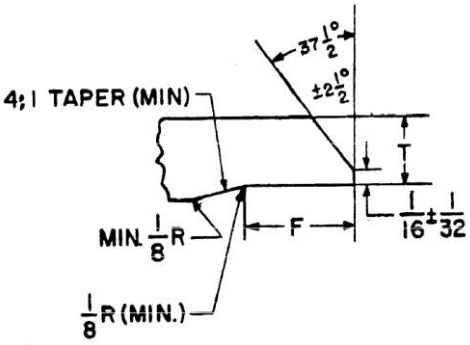
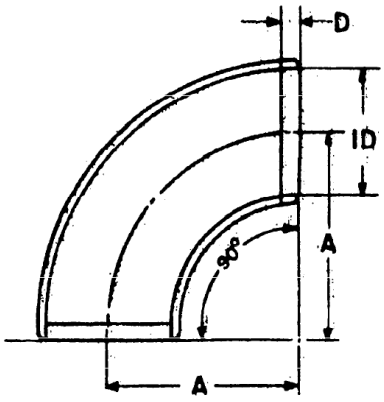


FIGURE 1. Prohibited weld locations.



NOTE: For other dimensions, see table I.

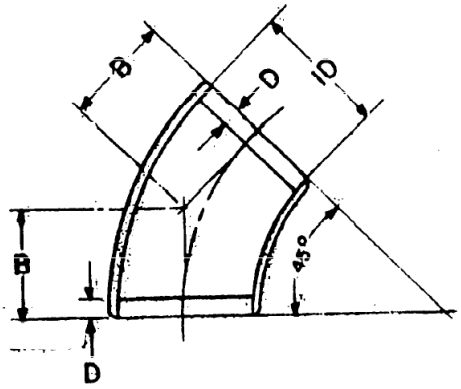
FIGURE 2. Edge preparation for welding.



NOTE: For dimensions and tolerances, see tables I and II.

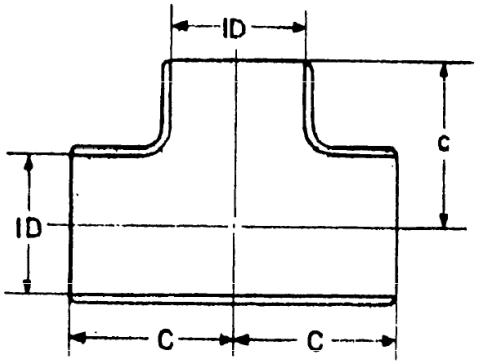
FIGURE 3. Ninety-degree elbow.

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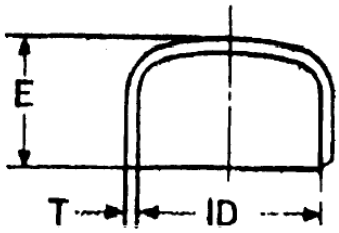
NOTE: For dimensions and tolerances, see tables I and II.

FIGURE 4. Forty-five-degree elbow.



NOTE: For dimensions and tolerances, see tables I and II.

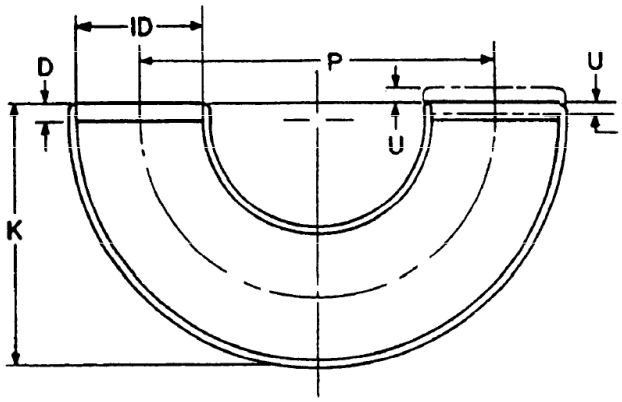
FIGURE 5. Straight tee.



NOTE: For dimensions and tolerances, see tables I and II.

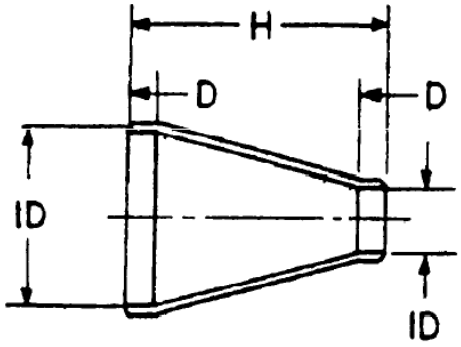
FIGURE 6. Cap.

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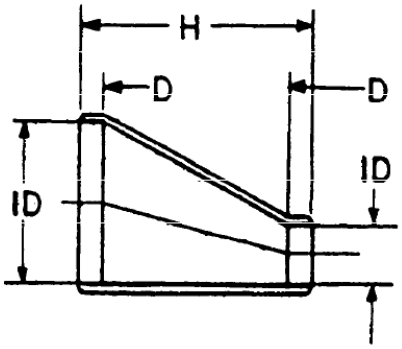
NOTE: For dimensions and tolerances, see tables I and II.

FIGURE 7. One-hundred-eighty-degree return bend.



NOTE: For dimensions and tolerances, see tables I, II, and III.

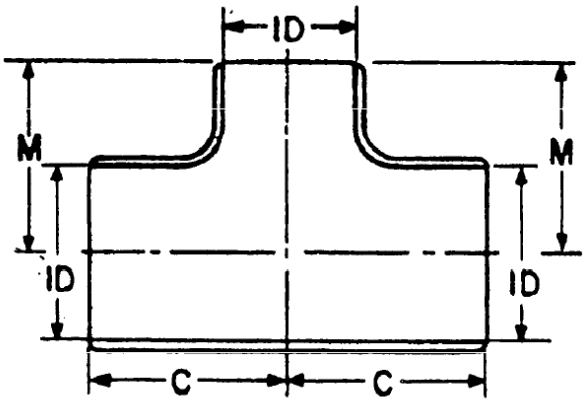
FIGURE 8. Concentric reducers.



NOTE: For dimensions and tolerances, see tables I, II, and III.

FIGURE 9. Eccentric reducers.

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NOTE: For dimensions and tolerances, see tables I, II, and IV.

FIGURE 10. Reducing tees.

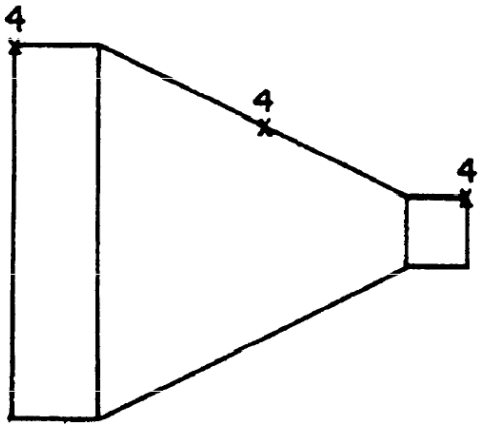


FIGURE 11. Concentric reducer inspection points.

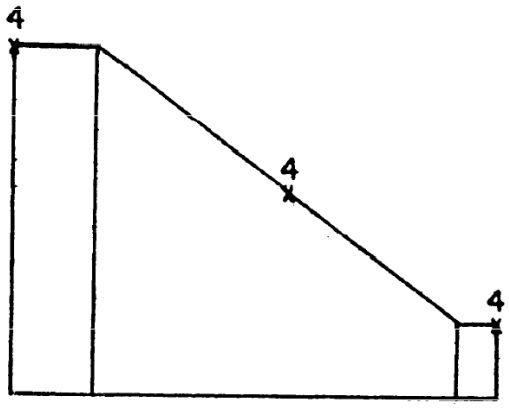


FIGURE 12. Eccentric reducer inspection points.

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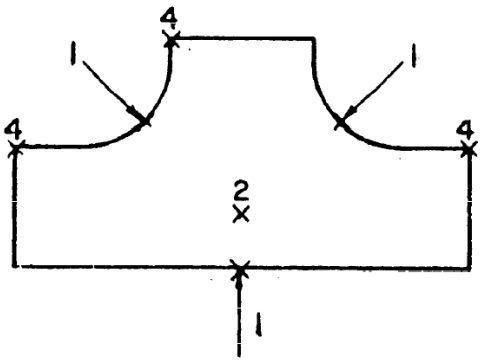


FIGURE 13. Tee inspection points.

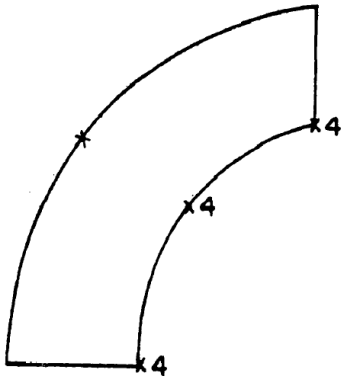


FIGURE 14. Ninety-degree elbow inspection points.

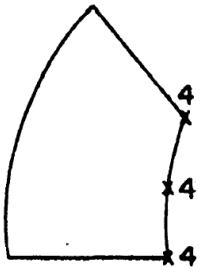


FIGURE 15. Forty-five-degree elbow inspection points.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy − SH
(Project 4730-2021-015)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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